

GenAI. Diffusion

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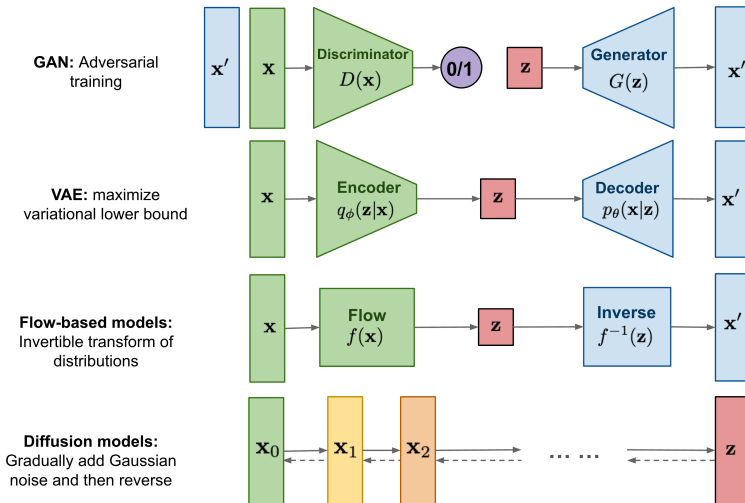
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Motivation



Introduction



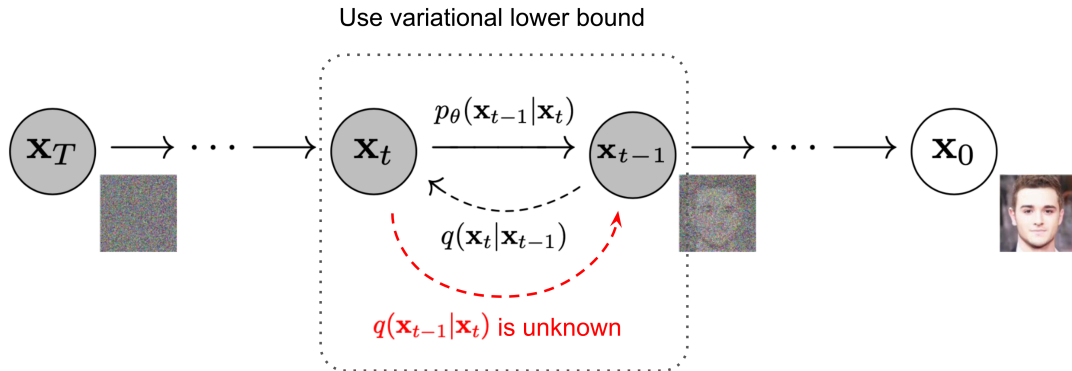
Motivation

- ▶ Unstable training process of GANs
- ▶ Blured and not realistic results of VAEs
- ▶ GANs and VAEs are overcomplicated (at least one extra network is used during the training process)
- ▶ Normalizing flows are limited due to fast Jacobian computation and invertibility

Diffusion models



Solution



Forward process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad (1)$$

Here, q is called *forward process* (like in NFs), or just *diffusion*, β 's are predefined on some grid (*variance schedule*). x_t is more noised than x_{t-1}

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1 - \alpha_t}\boldsymbol{\epsilon}_{t-1} && \text{;where } \boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\ &= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1 - \alpha_t\alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_{t-2} && \text{;where } \bar{\boldsymbol{\epsilon}}_{t-2} \text{ merges two Gaussians } (*). \\ &= \dots && (2) \\ &= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon} \\ q(\mathbf{x}_t|\mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \end{aligned}$$

Inverse process. Approximation

Problem: However, the inverse process $q(x_{t-1}|x_t)$ is **unknown**.

$$q(x_{t-1}|x_t) = \frac{q(x_t|x_{t-1}) * q(x_{t-1})}{\int q(x_t|x_{t-1}) * q(x_{t-1}) dx} \quad (3)$$

All we know that $q(x_t)$ and $q(x_t|x_{t-1})$ are Gaussian for all t (these distributions are called by Bayesians *conjugate*). Hence, $q(x_{t-1}|x_t)$ is also Gaussian!

Let's approximate then our unknown Gaussian denoising process $q(x_{t-1}|x_t)$ with neural network $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}|\mu_\theta(x_t), \sigma_\theta(x_t))$

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \quad p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)) \quad (4)$$

Inverse process. Variational Lower Bound

Problem: How to train p_θ ? Absolutely the same as with VAE!

$$\begin{aligned} -\log p_\theta(\mathbf{x}_0) &\leq -\log p_\theta(\mathbf{x}_0) + D_{\text{KL}}(q(\mathbf{x}_{1:T}|\mathbf{x}_0) \| p_\theta(\mathbf{x}_{1:T}|\mathbf{x}_0)) \\ &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_{\mathbf{x}_{1:T} \sim q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})/p_\theta(\mathbf{x}_0)} \right] \\ &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} + \log p_\theta(\mathbf{x}_0) \right] \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \end{aligned} \tag{5}$$
$$\text{Let } L_{\text{VLB}} = \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \geq -\mathbb{E}_{q(\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0)$$

Inverse process. Variational Lower Bound

Problem: How to train p_θ ? Absolutely the same as with VAE!

$$\begin{aligned} L_{\text{VLB}} &= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \\ &= \mathbb{E}_q \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \right) + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \quad (6) \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right] \\ &= \mathbb{E}_q \left[\underbrace{\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)}}_{L_T} + \sum_{t=2}^T \underbrace{\log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right] \end{aligned}$$

Inverse process. Variational lower bound

$$L_{\text{VLB}} = \mathbb{E}_q[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T}] + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}] \quad (7)$$

L_T is constant w.t. θ , $L_0 = \log \mathcal{N}(\mathbf{x}_0; \boldsymbol{\mu}_\theta(\mathbf{x}_1, 1), \boldsymbol{\Sigma}_\theta(\mathbf{x}_1, 1))$

Problem: $q(x_{t-1}|x_t, x_0)$ is **unkown**. Really?

Inverse process. Variational lower bound

$$\begin{aligned} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\ &\propto \exp \left(-\frac{1}{2} \left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t} \right) \right) \\ &= \exp \left(-\frac{1}{2} \left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) \mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0) \right) \right) \\ &= \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t) \end{aligned} \tag{8}$$

$$\begin{aligned} \tilde{\beta}_t &= \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \\ \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 \end{aligned} \tag{9}$$

Inverse process. Variational lower bound

Then,

$$L_{\text{VLB}} = C + \sum_{t=2}^T KL(\mathcal{N}(x_t; \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0, \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t) ||$$

$$\mathcal{N}(\mathbf{x}_t; \mu_{\theta}(x_t), \beta_{\theta}(x_t))) + \log \mathcal{N}(\mathbf{x}_0; \boldsymbol{\mu}_{\theta}(\mathbf{x}_1, 1), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_1, 1))(10)$$

DDPM



DDPM

According to (2),

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \quad (11)$$

, or, equivalently,

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\mathbf{I} * \epsilon_t | \epsilon_t \sim \mathcal{N}(0, 1) \quad (12)$$

Then, in (9)

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_t\right) \quad (13)$$

Moreover, let β_t be some constants

DDPM

... Then,

$$\begin{aligned} L_t &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{1}{2 \|\Sigma_\theta(\mathbf{x}_t, t)\|_2^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{1}{2 \|\Sigma_\theta\|_2^2} \left\| \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_t \right) - \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) \right\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{(1 - \alpha_t)^2}{2 \alpha_t (1 - \bar{\alpha}_t) \|\Sigma_\theta\|_2^2} \|\epsilon_t - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{(1 - \alpha_t)^2}{2 \alpha_t (1 - \bar{\alpha}_t) \|\Sigma_\theta\|_2^2} \|\epsilon_t - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, t)\|^2 \right] \end{aligned} \tag{14}$$

We can ignore the constant Σ , and then ...

DDPM

... all the training and inference processes become squeezed to

Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|^2$   
6: until converged
```

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

Relation to Langevin dynamics



Score Matching

Definition: Score matching aims to learn the gradients (termed *score*) of $\log p(\mathbf{x})$ with respect to $\mathbf{x}$ instead of directly modeling $\log p(\mathbf{x})$.

$$\mathcal{F}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x}) \quad (15)$$

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Optimization is done using the loss:

$$\mathcal{L}_{\text{matching}} = E_{\mathbf{x} \sim p(\mathbf{x})} \left[\text{tr}(\nabla_{\mathbf{x}} \mathcal{F}_\theta(\mathbf{x})) + \frac{1}{2} \|\mathcal{F}_\theta(\mathbf{x})\|_2^2 \right] \quad (16)$$

Langevin Dynamics and Diffusion Models

- ▶ Langevin dynamics is a stochastic process used to sample from a probability distribution by iteratively updating the state using gradient information and noise.
- ▶ It follows the equation:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \eta \nabla_{\mathbf{x}} \log p(\mathbf{x}_t) + \sqrt{2\eta} \mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(0, I) \quad (17)$$

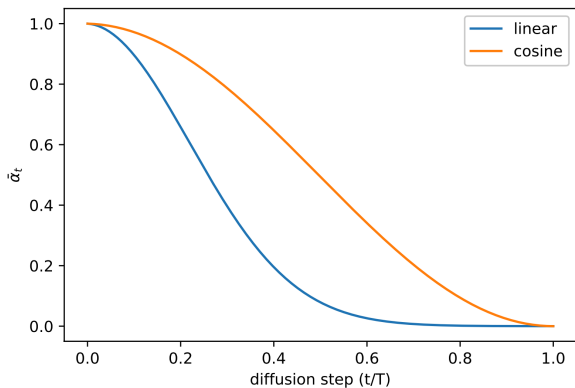
- ▶ Diffusion models can be seen as a discretization of Langevin dynamics, where the noise injection process allows for gradual refinement of samples.
- ▶ By applying a learned score function, diffusion models approximate the Langevin dynamics process to generate realistic samples.

Diffusion models now



Performance tricks. Variance scheduling

Set $\beta_t = \text{clip}(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, 0.999)$ $\bar{\alpha}_t = \frac{f(t)}{f(0)}$ where $f(t) = \cos\left(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\right)^2$ instead of gradually increasing β_t from 10^{-4} to 0.02



Performance tricks

- ▶ Learn the variance. Set $\Sigma_{\theta}(\mathbf{x}_t, t) = \exp(\mathbf{v} \log \beta_t + (1 - \mathbf{v}) \log \tilde{\beta}_t)$ instead of $\Sigma_{\theta}(\mathbf{x}_t, t) = \sigma_t^2 \mathbf{I}$ in DDPM

- ▶ Sample every S steps (strided sampling)

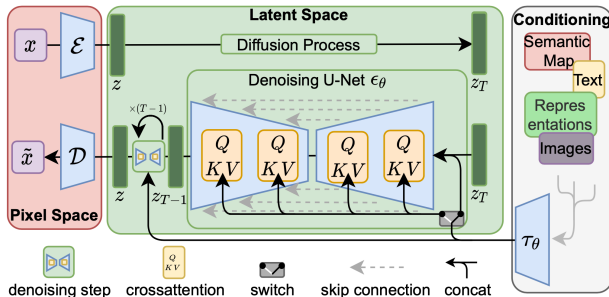
$$q_{\sigma, \tau}(\mathbf{x}_{\tau_{i-1}} | \mathbf{x}_{\tau_t}, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{\tau_{i-1}}; \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2 \frac{\mathbf{x}_{\tau_i} - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}}, \sigma_t^2 \mathbf{I})$$

Latent Diffusion

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right) \cdot \mathbf{V}$$

where $\mathbf{Q} = \mathbf{W}_Q^{(i)} \cdot \varphi_i(\mathbf{z}_i)$, $\mathbf{K} = \mathbf{W}_K^{(i)} \cdot \tau_\theta(y)$, $\mathbf{V} = \mathbf{W}_V^{(i)} \cdot \tau_\theta(y)$ (18)

and $\mathbf{W}_Q^{(i)} \in \mathbb{R}^{d \times d_\epsilon^i}$, $\mathbf{W}_K^{(i)}, \mathbf{W}_V^{(i)} \in \mathbb{R}^{d \times d_\tau}$, $\varphi_i(\mathbf{z}_i) \in \mathbb{R}^{N \times d_\epsilon^i}$, $\tau_\theta(y) \in \mathbb{R}^{M \times d_\tau}$



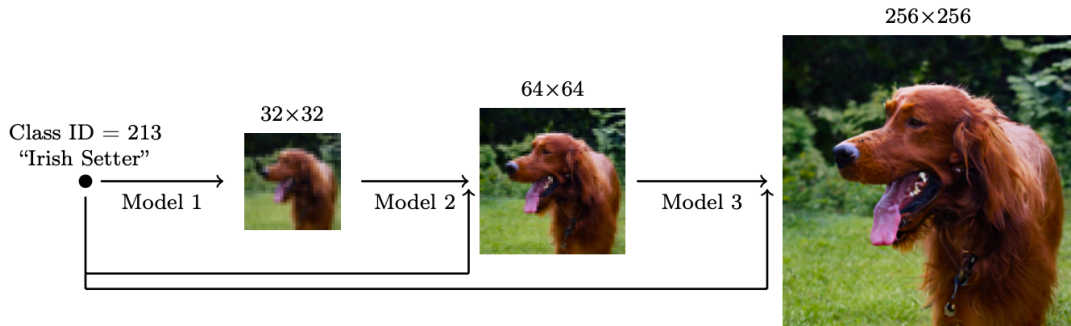
Classifier guidance

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t, y) &= \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log q(y|\mathbf{x}_t) \\ &\approx -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) + \nabla_{\mathbf{x}_t} \log f_{\phi}(y|\mathbf{x}_t) \\ &= -\frac{1}{\sqrt{1-\bar{\alpha}_t}} (\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) - \sqrt{1-\bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log f_{\phi}(y|\mathbf{x}_t))\end{aligned}\tag{19}$$

Classifier-Free Guidance

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|y) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) \\ &= -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \left(\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) \\ \bar{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t, y) &= \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - \sqrt{1-\bar{\alpha}_t} w \nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) \\ &= \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) + w(\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)) \\ &= (w+1)\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - w\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\end{aligned}\tag{20}$$

Cascaded diffusion



Thank you for your attention!



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